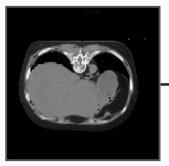
## A PLC-Net generator (2-D U-Net)



**NCCT x**
1 × 128 × 128
HU [–200, 400] → [0, 1]

2.5-D variant: 11 adjacent slices as input channels (in = 11)

**enc1** 1 → 64 128²
[Conv3×3 → GN → LReLU] ×2

MaxPool 2×2

**enc2** 64 → 128 64²
[Conv3×3 → GN → LReLU] ×2

MaxPool 2×2

**enc3** 128 → 256 32²
[Conv3×3 → GN → LReLU] ×2

MaxPool 2×2

**enc4** 256 → 512 16²
[Conv3×3 → GN → LReLU] ×2

MaxPool 2×2

**Bottleneck** 512 → 512 8²
[Conv3×3 → GN → LReLU → Drop] ×2

*Encoder is unconditioned: shared across phases & levels*

copy & concat

copy & concat

copy & concat

copy & concat

**dec4** (256+512) → 256 16²
Conv3×3 → GN → LReLU → Drop
Conv3×3 → GN → LReLU

**dec3** (128+256) → 128 32²
Conv3×3 → GN → LReLU → Drop
Conv3×3 → GN → LReLU

**dec2** (64+128) → 64 64²
Conv3×3 → GN → LReLU → Drop
Conv3×3 → GN → LReLU

**dec1** (64+64) → 64 128²
Conv3×3 → GN → LReLU → Drop
Conv3×3 → GN → LReLU

ConvT 2×2, s2 · 512 → 256

ConvT 2×2, s2 · 256 → 128

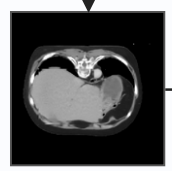
ConvT 2×2, s2 · 128 → 64

ConvT 2×2, s2 · 64 → 64

**FiLM 1** **FiLM 2** **FiLM 3** **FiLM 4** **FiLM 5**

γ, β

**Conv1×1**
64 → 1
→ Sigmoid



**Synthetic CECT ŷ**
1 × 128 × 128

ŷ

**GN** = GroupNorm (8 groups) · **LReLU** = LeakyReLU(0.2) · **Drop** = Dropout(0.2) · **FiLM**: y = x·(1+γ(c)) + β(c)

c (detached: no gradient to G)

## B Conditioning branch (decoder only)

**Phase id p**
venous = 0 · arterial = 1

**nn.Embedding**(2, 64)

phase emb.

**Target organ levels z ∈ ℝ⁸**
aorta
portal / splenic vein
inferior vena cava
heart
liver
pancreas
gallbladder
colon

**Linear**(8 → 64)
zero-init

level emb.

*z = median HU of each organ in the target CECT (TotalSegmentator mask), z-scored with train-split statistics pooled over phases; missing organ → 0*

**Train:** z from the real CECT (oracle)

**Inference:** population_phase (per-phase train mean) · fixed_z · oracle

**c** ∈ ℝ⁶⁴

**FiLM MLP** (one per site, ×5)
Linear(64 → h) → ReLU → Linear(h → 2C)
last layer zero-init → (γ, β) ∈ ℝ^C × ℝ^C

**FiLM**
y = x · (1 + γ) + β
γ = β = 0 at init
⇒ identity at step 0

| site | after | C | h |
|---|---|---|---|
| FiLM 1 | bottleneck | 512 | 128 |
| FiLM 2 | dec4 | 256 | 64 |
| FiLM 3 | dec3 | 128 | 64 |
| FiLM 4 | dec2 | 64 | 64 |
| FiLM 5 | dec1 | 64 | 64 |

## C Conditional PatchGAN D

*training only*

**NCCT x** **real CECT y**

**concat** [ x , y or ŷ ] 2 × 128²
*D(x, ·) judges the pair, not ŷ alone*

**Conv4×4, s2** 2 → 64 → LReLU
no norm · 64²

**Conv4×4, s2** 64 → 128 → BN → LReLU
32²

**Conv4×4, s2** 128 → 256 → BN → LReLU
16²

**Conv4×4, s1** 256 → 512 → BN → LReLU
15²

**Linear**
(64 → 512)
zero-init

bias

**Conv4×4, s1** 512 → 1
14 × 14 patch logit map (real / fake)

$L_{LSGAN}$

feature matching

## D Training objective

**TotalSegmentator organ mask M**
*training only*

**Target CECT y**
(venous or arterial)

$L_G = \lambda_{L1}(t) \cdot \|\hat{y} - y\|_1 + 20 \cdot L_{organ} + \lambda_{adv}(t) \cdot L_{LSGAN} + 10 \cdot L_{FM}$

$L_D = ½ [ (D(x, y) - 1)^2 + D(x, \hat{y})^2 ]$ (LSGAN)

- $\lambda_{L1}(t)$: 100 → 25, linear decay over epochs 10–30
- $L_{organ}$: organ-weighted L1, weights from M — aorta / IVC / portal v. 6, heart & great vessels 4, liver 3, other solid organs 2, background 0.5, lung / bone / muscle 0.25, GI tract 0
- $\lambda_{adv}(t)$ = 0.5, linear warm-up over 5 epochs
- $L_{FM}$: L1 between D features of real and synthetic pairs (4 blocks)
- Adam (β = 0.5, 0.999), $lr_G$ $2 \cdot 10^{-4}$, $lr_D$ $1 \cdot 10^{-4}$, batch 16, ≤ 70 epochs (early stop, patience 30)